



AV security whitepaper

An outline of all the potential security risks of autonomous vehicles, including motives, vectors and impact. <https://digit.in/oct19-66>



Cyberphysical risks

What are the security scenarios of vehicles that are connected to each other, and all get hacked at once? <https://digit.in/oct19-66>

prints will expand. In short, the cities are expected to grow bigger.

CYBERSECURITY

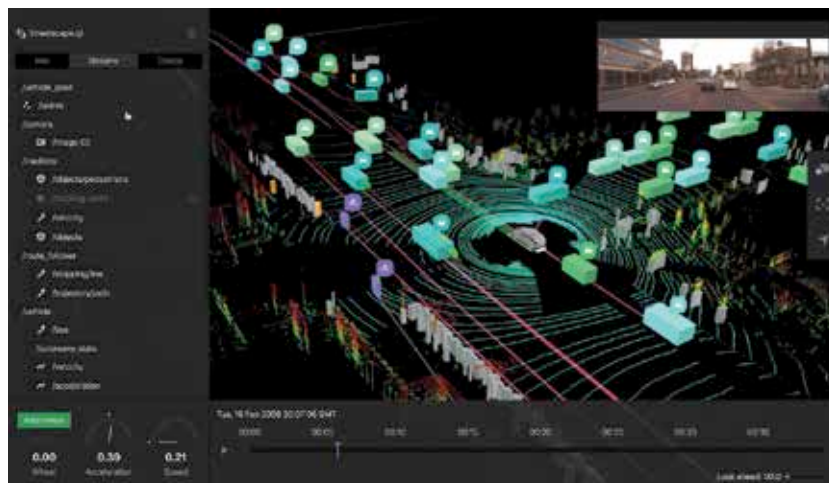
As level 5 autonomous cars are not here yet, researchers have very little direct experience of cybersecurity for autonomous cars. However, security researchers have outlined some potential scenarios so that adequate safeguards can be taken. Some of these threats may not be entirely new, but can be much more scary in the context of self driving cars. This is because computers, smartphones, and even cloud storage systems need to be safeguarded for only the data they have on them. In a sense, what makes these valuable is entirely in cyberspace. However, when it comes to autonomous cars, human lives can be at threat.

Take ransomware for example. It infects a machine, and prevents a user from accessing the contents unless a payment is made. Now imagine getting into a car, the doors close and lock up. You cannot exit or move anywhere, unless you make a cryptocurrency payment. Even the safety mechanisms built into the vehicle can be counterproductive. In fully-autonomous cars, there might not be a fall back for a driver to take over, in case anything goes wrong, and even if there is, there might not be anyone in the vehicle capable of actually controlling the vehicle. Automobile manufacturers anticipate solving such problems by configuring safe destinations for the vehicle to navigate to, in case there is any problem, such as it senses that it has been hacked. However, what if this mechanism kicks in during an emergency, such as a visit to a hospital.

According to researchers from the University of Michigan, the security threats facing autonomous vehicles are just not very well understood. This is because autonomous cars are both physical and virtual systems. That means the threat exposure is double. They are vulnerable to traditional carjackers and thieves. They

are also vulnerable to hackers, spoofers and ddos attacks. The researchers have developed a tool to identify the motivations of various kinds of threats, the vector of attack, and the impacts. Andre Weimerskirch who leads University of Michigan's cybersecurity working group says, "Cybersecurity is an overlooked area

Georgia Tech are running simulations to understand what it would be like if hackers unleash mass mayhem, by taking over multiple vehicles in a city. To gridlock Manhattan for example, the hackers only have to make 20 percent of the vehicles stop in a location that blocks the traffic. Now, the figure holds true irrespective of how many



How the Uber AVS visualises its environment

of research in the development of autonomous vehicles. Our tool marks not only an important early step in solving these problems, but also presents a blueprint to effectively identify and analyse cybersecurity threats and create effective approaches to make autonomous vehicle systems safe and secure."

Now, smart autonomous vehicles are likely to have systems on board to interact with the smart cities around them. These include two-way communications with toll booths, other vehicles, signals, recharging stations, and sensors in the home. All of these are potentially vectors of attack. Apart from individual vehicles, hackers can also target a number of vehicles at once. For example, hacktivists may want to take all the vehicles of a particular manufacturer offline, for ideological reasons. Some potential attacks can be even more scary. Cyberterrorists for example, potentially grid lock an entire city by taking control of a number of vehicles. Researchers from

cars on the roads are autonomous or driven by people. This scenario can come to pass even before roads are populated by fully autonomous cars alone. One of the researchers, David Yanni explains, "randomly stalling 20 percent of cars during rush hour would mean total traffic freeze. At 20 percent, the city has been broken up into small islands, where you may be able to inch around a few blocks, but no one would be able to move across town." Now Manhattan has well-ordered roads that make the flow of traffic smooth. Factoring in the panic following such an attack, and bad road networks in other cities, the percentage of vehicles needed to lock up a city could be far fewer. The researchers have also come up with some methods to prevent such gridlock attacks, including making sure that as many vehicles are isolated from each other. If by design, the networks that control autonomous cars do not allow multiple vehicles to be commandeered at once, then that is a safety feature. 